

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of BSc (Biological sciences) Examination April-May, 2019

BS209: Animal Sciences

Date: 29/04/2019 Day: Monday Time: 10.00 a.m. To 10.30 a.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
-

Q - I	Choose the correct answer for the following questions.	20
1.	Binary fission is the characteristic feature of reproduction in (i) Protozoa (ii) yeast (iii) both i and ii (iv) neither i nor ii	
2.	Grid cells and place cells are found in which region of the brain? (i) hypothalamus (ii) spinal cord (iii) hippocampus (iv) None of these	
3.	The biological species concept is based on (i) Evolutionary lineage (ii) Reproductive isolation (iii) Unique morphological and molecular characteristics (iv) Unique ecological niche	
4.	The famous book 'Origin' of Species' was written by (i) Jean Baptiste Lamarck (ii) Weismann (iii) Charles Darwin (iv) None of these	

5.	Which of the following is not a mollusc? (i) Octopus (ii) Sea anemone (iii) Snail (iv) Oyster	
6.	Club-shaped protuberances (bulging) that appear knob-like structures and make synaptic connections with other cells are called Boutons. They contain several small vesicles called Synaptic Vesicles that contain (i) peptides (ii) acetylcholine (iii) neurylcholine (iv) All of these	
7.	Crocodiles, Birds and Mammals have four chambered heart. ☐ True ☐ False	
8.	Which of the following animals does not show external fertilization (i) Frog (ii) Rohu Fish (iii) Rat (iv) Sea urchin	
9.	Territoriality is the behavior of some animals to defend areas for their reproduction and resources. The main role of the territoriality is (i) mating and rearing young ones (ii) to ensure food supply (iii) to reduce competition (iv) all of the above	
10.	Fibroblast is a component of (i) Connective tissue (iii) Epithelial Tissue (ii) Muscular Tissue (iv) Nervous Tissue	
11.	Platelets and RBCs possess (i) Bi-lobed nucleus (ii) Tri-lobed nucleus (iii) Multi-lobed nucleus (iv) None of the above	
12.	Unfertilized ovum is discarded along with degenerated endometrium during menstrual phase of menstrual cycle. ☐ True ☐ False	
13.	The term which is used for learning that does not respond to stimuli (i) Imprinting (ii) Sensation (iii) Habituation iv) Sensitization	
14.	Animals standing between two groups of animals are called --- (i) joining links (ii) connecting links (iii) intermediate links (iv) none of these	
15.	Kangaroos have a pouch –like structure for nurturing and feeding their babies till they are mature. The structure is called as (i) marsupium (ii) brood (iii) nursery (iv) none of these	

16.	Transitional epithelium forms the lining of the urinary bladder. ☐True ☐False	
17.	Spermatogenesis has long resting periods compared to oogenesis. ☐True ☐False	
18.	Incomplete C-shaped cartilaginous rings allow the trachea to collapse slightly when food passes down the esophagus. ☐True ☐False	
19.	Structural similarity between various organs of different organisms demonstrate (i) Analogy (ii) Homology (iii) Heterology (iv) All the above	
20.	Pseudostratified epithelium contains (i) One cell layer of same length (ii) One cell layer of dissimilar length (iii) Many cell layers of same length (iv) Many cell layers of varying length	

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of BSc (Biological sciences) Examination April-May, 2019

BS209: Animal Sciences

Date: 29/04/2019 Day: Monday Time: 10.30 a.m. To 1.00 p.m. Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II	Answer the following questions as directed	20
	Answer <u>any four</u> of the following:	
1.	Write down the basis of animal classification. What do you understand by the terms Systematics, taxonomy and phylogeny in animal classification?	
2.	Draw a well labeled diagram of human spermatozoan and discuss the spermatogenesis process in human beings.	
3.	Briefly explain how chondrichthyes fishes are different from osteichthyes fishes? Comment upon the habitats and external features of animals belonging to class Aves.	
4.	How is operant conditioning different from classical conditioning in associative learning? Explain with examples.	
5.	What is deglutition? Discuss the various phases of food digestion.	
SECTION - II		
Q-III	Answer the following questions as directed	30
	Answer <u>any six</u> of the following:	
1.	Discuss about the various types of Epithelial tissues found in humans.	
2.	State Hardy-Weinberg law and discuss the evolutionary forces that may disturb the Hardy-Weinberg equilibrium.	
3.	What is Vasectomy? Diagrammatically describe male reproductive system.	
4.	Elaborate several factors that are responsible for animal evolution.	
5.	Mention various components of respiratory system and their role during respiration.	
6.	Define 'animal behaviour' and explain 'parental care' in different animals.	
7.	Diagrammatically compare acoelomate, pseudocoelomate and coelomate animals with appropriate examples.	